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0625/43

May/June 2019

1 hour 15 minutes

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

DO **NOT** WRITE IN ANY BARCODES.

Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m/s^2).

The number of marks is given in brackets [] at the end of each question or part question.

This syllabus is regulated for use in England, Wales and Northern Ireland as a Cambridge International Level 1/Level 2 Certificate.

This document consists of **18** printed pages and **2** blank pages.

- 1 Fig. 1.1 shows a distance-time graph for a cyclist travelling between points P and V on a straight road.

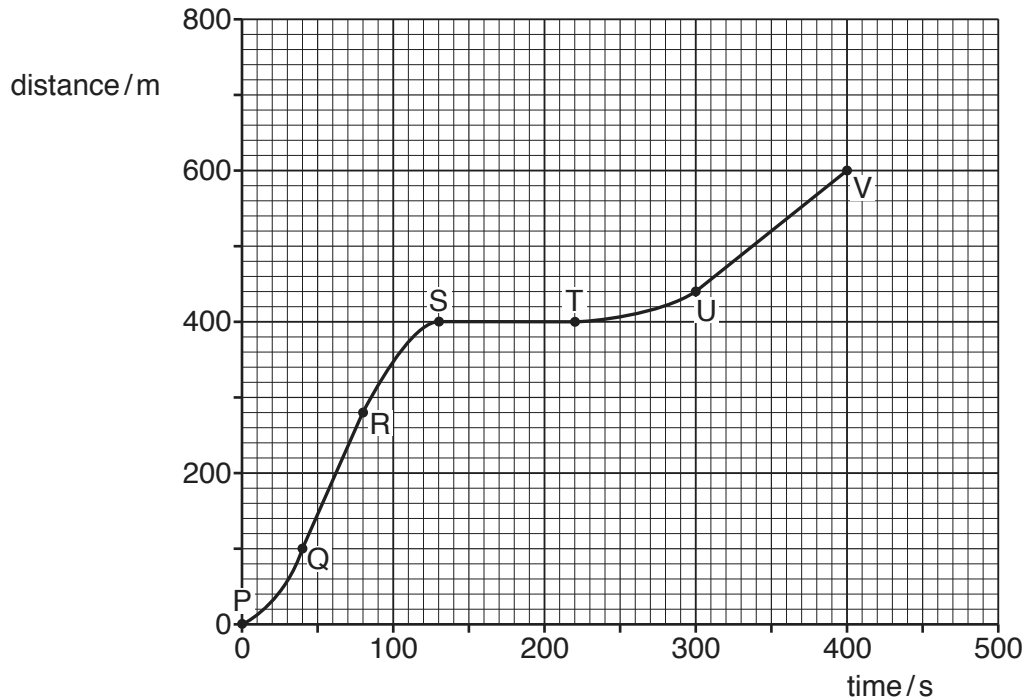


Fig. 1.1

- (a) Describe the motion between:

Q and R

R and S

S and T.

[3]

- (b) Calculate the speed between U and V.

speed = [2]

- (c) After point V, the straight road continues down a steep hill. The cyclist travels down the steep hill. He does not apply the brakes and all resistive forces can be ignored.

On Fig. 1.1, sketch a possible motion for the cyclist after V.

[1]

[Total: 6]

- 2 Fig. 2.1 is the top view of a small ship of mass $1.2 \times 10^6 \text{ kg}$. The ship is moving slowly sideways at 0.040 m/s as it comes in to dock.

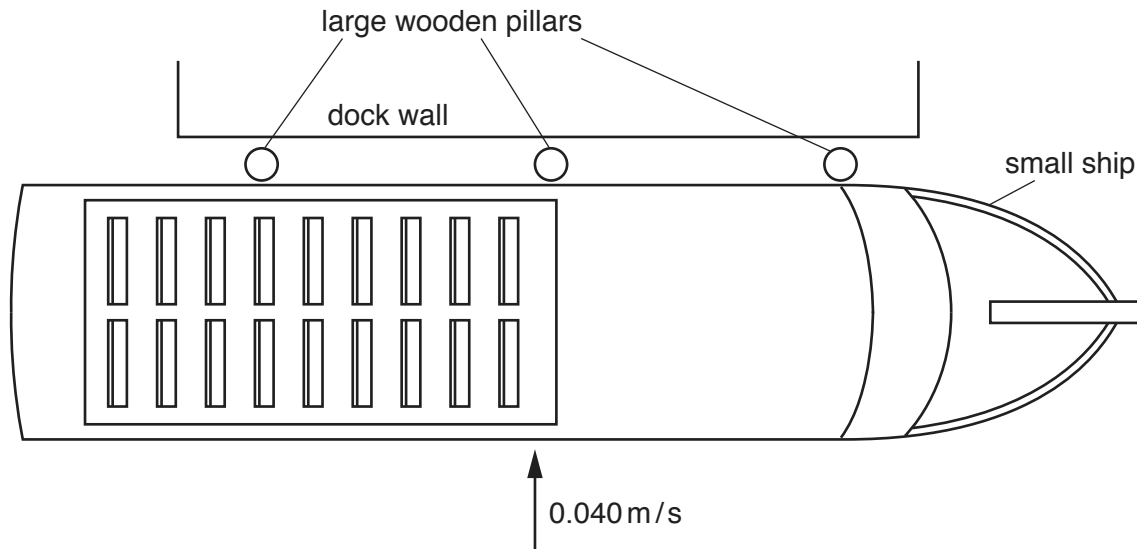


Fig. 2.1

The ship hits the wooden pillars which move towards the dock wall.

- (a) Calculate the kinetic energy of the ship before it hits the pillars.

kinetic energy = [2]

- (b) The ship is in contact with the pillars for 0.30 s as it comes to rest.

Calculate the average force exerted on the side of the ship.

force = [4]

- (c) Assume that the kinetic energy calculated in (a) is used to do work moving the pillars.

Calculate the distance moved by the pillars.

distance = [2]

- (d) Dock walls sometimes have the pillars replaced with rubber car tyres.

Explain how this reduces the possibility of damage when a boat docks.

.....
.....
..... [1]

[Total: 9]

- 3 Fig. 3.1 shows a small submarine submerged below the surface of the sea.

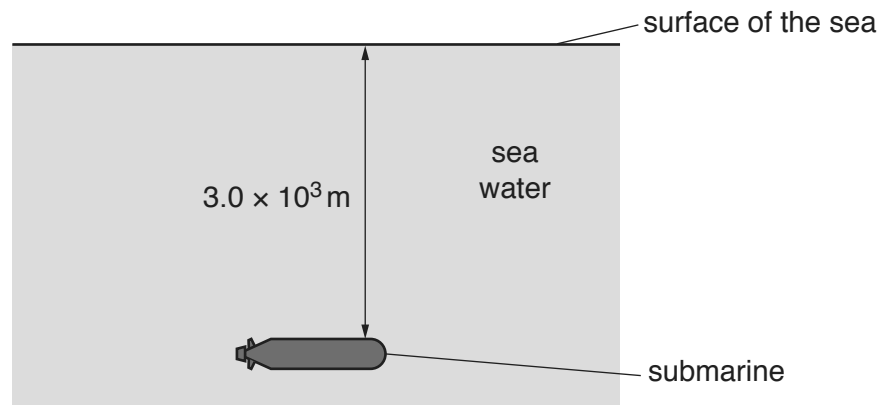


Fig. 3.1

- (a) The density of sea water is 1030 kg/m^3 .

Calculate the pressure due to the sea water on the top of the submarine when it is $3.0 \times 10^3 \text{ m}$ below the surface.

pressure = [2]

- (b) The submarine emits a pulse of sound to detect other objects in the sea. The speed of sound in sea water is 1500 m/s. An echo is received with a time delay of 0.50 s after the original sound is emitted.

(i) Calculate the distance between the submarine and the other object.

distance = [3]

(ii) Another pulse of sound is emitted through the air when the submarine is on the surface.

An echo is received from a second object that is in the air. This echo is received 0.50 s after the pulse of sound is emitted.

Compare the distance of the second object from the submarine with the distance calculated in (b)(i). Tick **one** box. Give a reason for your answer.

distance is smaller

☐

distance is the same

☐

distance is larger

☐

Reason [1]

[Total: 6]

- 4 (a) Water molecules escape to the atmosphere from water boiling in a pan. Water molecules evaporate from the surface of a bowl of cool water and also escape to the atmosphere.

State **two** ways in which boiling is different from evaporation.

1.

 2.

[2]

- (b) Fig. 4.1 shows a heater in a metal block.

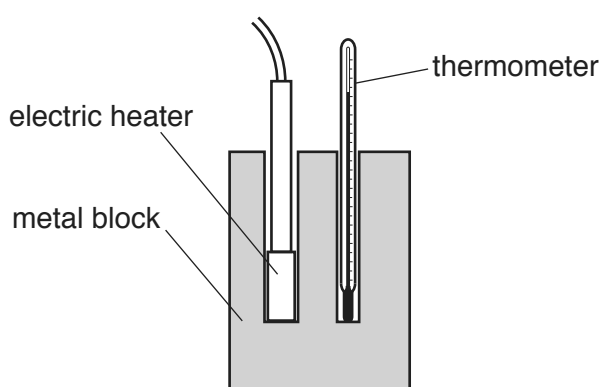


Fig. 4.1

The power of the heater is 370 W and it is switched on for 4.0 minutes. The metal block has a specific heat capacity of $420 \text{ J/(kg } ^\circ\text{C)}$ and a mass of 5.0 kg.

Calculate the increase of temperature of the block. Assume all the thermal energy from the heater is transferred to the block.

temperature increase = [4]

[Total: 6]

- 5 Fig. 5.1 shows a cross-section of the inside of a vacuum flask containing a cold liquid. The walls of the vacuum flask are made of glass.

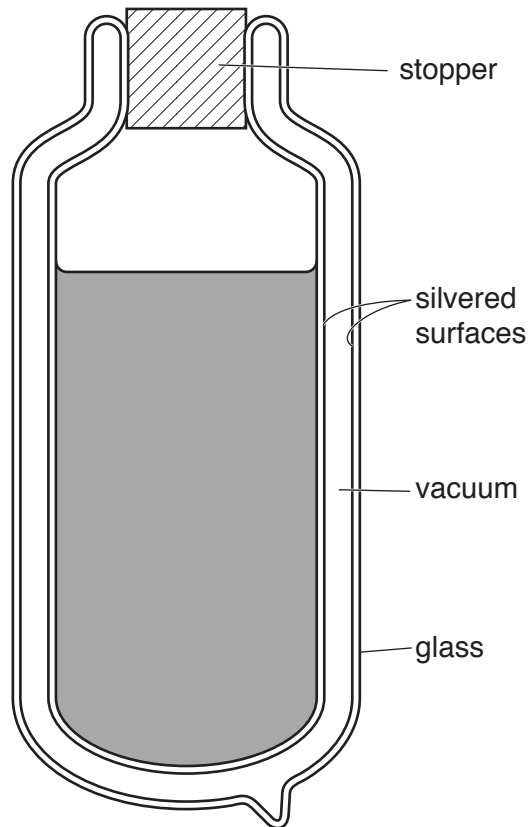


Fig. 5.1

- (a)** The vacuum flask is being used to keep a liquid cool on a hot day.

Explain how the labelled features of the vacuum flask keep the liquid cool by reducing thermal energy transfer. Include the names of the processes involved.

.....

.....

.....

.....

.....

.....

..... [5]

- (b)** Suggest a suitable material for the stopper.

..... [1]

[Total: 6]

- 6 (a) Fig. 6.1 shows wavefronts of a wave approaching a narrow gap and passing through the gap. The wavelength is λ .

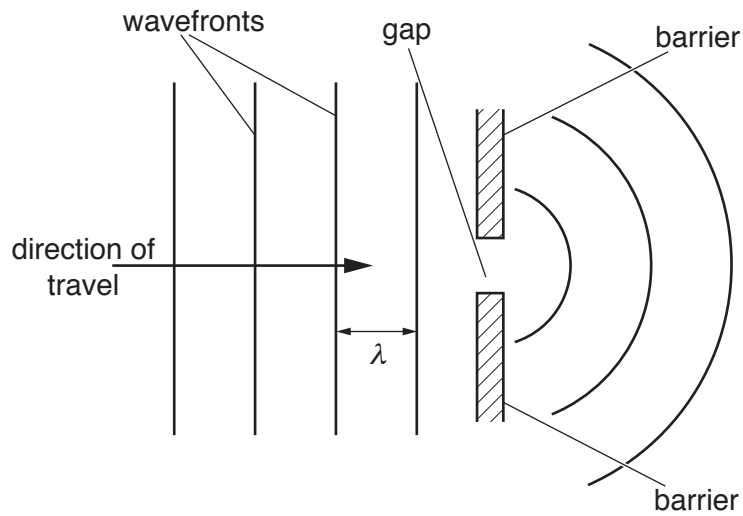


Fig. 6.1

- (i) State the name of the process that occurs as the wave passes through the gap.

..... [1]

- (ii) A wave with a wavelength $\frac{\lambda}{2}$ approaches the same gap.

On Fig. 6.2, draw three wavefronts for this wave as it approaches the gap and three more wavefronts as the wave continues beyond it. [3]



Fig. 6.2

- (b) Table 6.1 shows 5 different types of electromagnetic wave.

In the blank column in Table 6.1, write the numbers 1 to 5 to show the order of wavelength. Write 1 for the wave with the shortest wavelength and 5 for the wave with the longest wavelength. [2]

Table 6.1

type of electromagnetic wave	order of wavelength
gamma rays	
light	
microwaves	
ultraviolet	
X-rays	

- (c) (i) State the speed of radio waves in air.

..... [1]

- (ii) A radio station transmits radio waves with a frequency of 96 MHz. Calculate the wavelength of these radio waves.

wavelength = [3]

[Total: 10]

- 7 Fig. 7.1 shows light approaching a boundary between two materials at speed v . The speed of the light after crossing the boundary is $1.3v$.

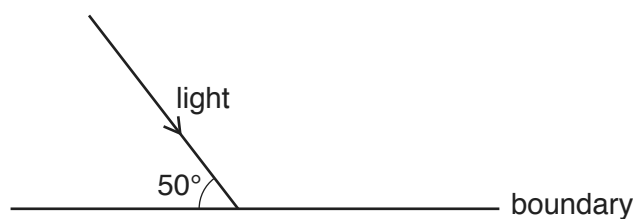


Fig. 7.1

- (a) Determine the angle of incidence.

angle of incidence = [1]

- (b) Calculate the angle of refraction.

angle of refraction = [3]

[Total: 4]

- 8 Fig. 8.1 shows a 240 V mains supply connected to an air-conditioning unit and a freezer. A fuse X is placed in the circuit as shown.

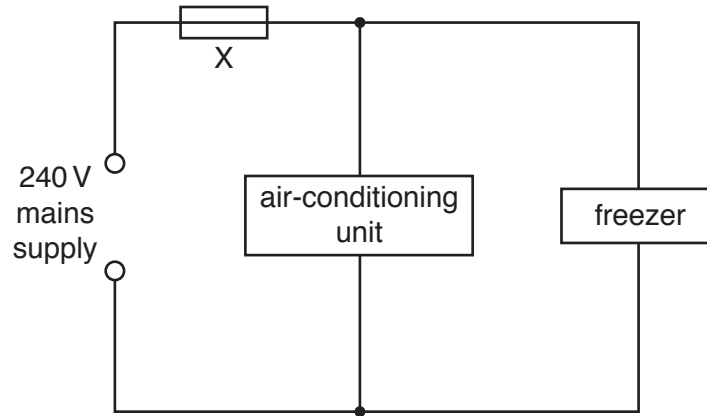


Fig. 8.1

The freezer has an operating power of 700 W.

- (a) Calculate the current in the freezer.

current = [2]

- (b) The maximum operating current of the air-conditioning unit is 7.5 A.

Fuses of current rating 1 A, 3 A, 5 A, 10 A, 13 A and 30 A are available.

Suggest a suitable rating for fuse X. Give **two** reasons for your answer.

fuse rating

Reason 1

.....

.....

Reason 2

.....

..... [3]

- (c) A fuse is made out of a short length of wire.

Explain why fuses of a higher rating are made of thicker wire.

.....

.....

.....

.....

..... [3]

- (d) Electrical energy can be obtained from renewable and non-renewable sources of energy.

- (i) State **two** renewable sources of energy.

Source 1

Source 2 [2]

- (ii) State **one** social, economic or environmental disadvantage of one of your answers to (d)(i).

.....

.....

..... [1]

[Total: 11]

- 9 (a) Fig. 9.1 shows an electrical component.

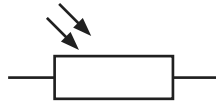


Fig. 9.1

State the name of the component shown in Fig. 9.1. [1]

- (b) In the space below, write down the truth table for a NOR gate.

[2]

(c) Fig. 9.2 shows the connections between two logic gates.

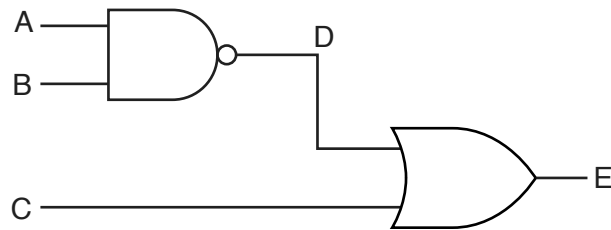


Fig. 9.2

Complete the truth table shown in Table 9.1 for this combination of logic gates.

Table 9.1

inputs			intermediate point	output
A	B	C	D	E
0	1	1		
1	0	1		
1	1	0		
1	1	1		

[3]

(d) Referring to a simple electron model, state what distinguishes electrical conductors from electrical insulators.

.....

.....

.....

..... [1]

[Total: 7]

10 Fig. 10.1 shows a simple alternating current generator.

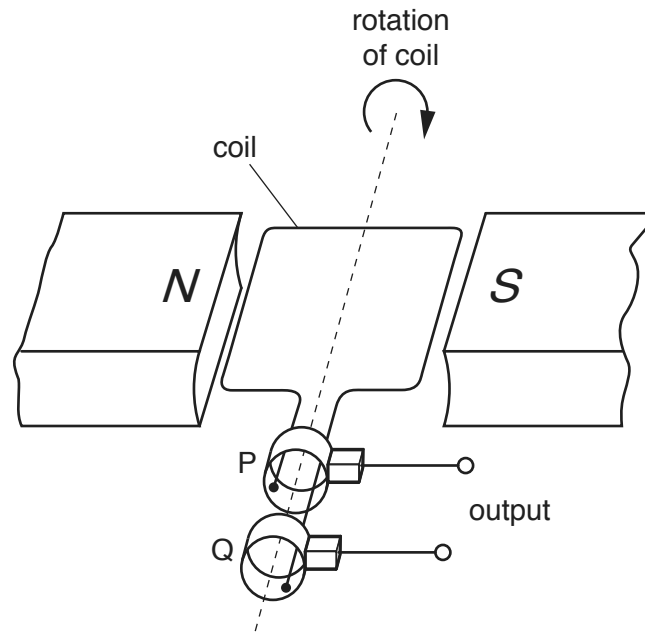


Fig. 10.1

- (a) On Fig. 10.2, sketch a graph to show how the electromotive force (e.m.f.) induced varies with time for one revolution of the coil. Assume that the coil starts in the horizontal position, as shown in Fig. 10.1. Label the points on the time axis where the coil has completed $1/4$ revolution and $3/4$ revolution. [3]

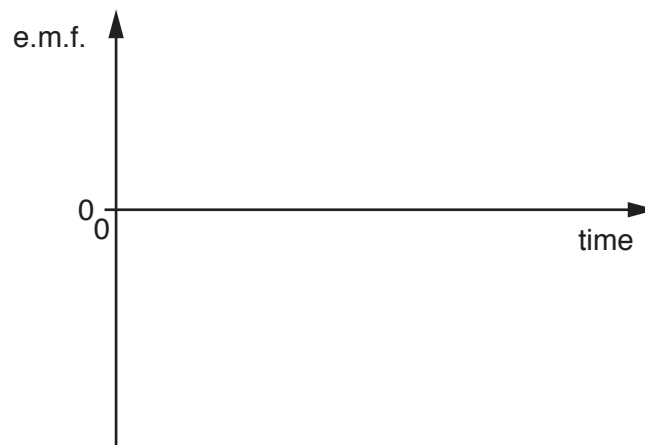


Fig. 10.2

- (b) Explain why an e.m.f. is induced only when the coil is turning.

.....

.....

..... [1]

- (c) State the name of the components labelled P and Q and state their purpose.

Name:

Purpose:

..... [2]

- (d) State **two** possible changes that cause a larger e.m.f. to be induced.

1.

2. [2]

[Total: 8]

- 11 (a)** Americium (Am) is a radioactive isotope. A nucleus of americium contains 95 protons and 146 neutrons. It decays by emitting an α -particle to form a nucleus of an isotope of neptunium (Np).

Write down the nuclide equation for the decay of americium to neptunium.

[4]

- (b)** Ionisation smoke detectors contain americium and two small electrodes with a small voltage between them. The air between the electrodes is ionised by α -particles so that there is a small electric current between the electrodes.

- (i)** Suggest and explain the effect of smoke on the current between the electrodes in the smoke detector.

Suggestion:

Explanation:
 [1]

- (ii)** Suggest **two** reasons for using an α -particle emitter in a smoke detector.

Reason 1

Reason 2
 [2]

[Total: 7]

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